

Sunin Kim

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OBJECTIVE

My goal is to integrate robots into everyday life, enabling people to lead enriched lives by offloading tedious tasks and enjoying meaningful interactions with robots.

EXPERIENCE

- **CarbonSix** Nov. 2025 - Present
Robot AI researcher
 - Conducted research on physical intelligence through human motion.
 - Created a simulation benchmark to evaluate model performance and generated data.
 - Researched to improve performance from deployment using reinforcement learning.
- **NAVER LABS** Feb. 2022 - Nov. 2025
Robot control researcher
 - Conducted research on general manipulation policies using vision-language models (VLMs).
 - Developed optimal trajectory planning algorithms for mobile robot path following.
 - Researched unsupervised algorithms for safe skill discovery in diverse manipulation tasks.

EDUCATION

- **Korea University** Mar. 2020 - Feb. 2022
M.S. in Department of Mechanical Engineering, Advisor: Jae-Bok Song
Thesis - Hierarchical object manipulation system for procedural unit task learning
Seoul, South Korea
- **Korea University** Mar. 2014 - Feb. 2020
B.S. in Department of Mechanical Engineering
Seoul, South Korea

PROJECTS

- **Learning from deployment** Nov. 2025 - Present
On-site training following pre-training
 - Behavioral improvements through human intervention during inference.
 - Learning from failure and reinforcement learning.
- **Locomotion and Manipulation** Feb. 2025 - Nov. 2025
Learning-based control for embodied robots
 - Developed data pipelines for training and fine-tuning vision-language action models (VLAs).
 - Implementing diffusion-based manipulation policies on large-scale datasets.
 - Generated motions in simulation and transferred to real-world robots.
- **2nDC Robot** Jul. 2022 - Feb. 2025
Mobile robots for data center
 - Developed a docking algorithm for autonomous cart loading and charging.
 - Implemented a time-optimal path parameterization algorithm for efficient trajectory planning.
 - Developed a marker-based localization algorithm using sensor fusion with an Extended Kalman Filter (EKF).
- **AMBIDEX** Feb. 2022 - Jul. 2022
Cable-driven light weight bi-manipulation robot
 - Conducted research on safe and efficient control strategies.
 - Explored skill discovery algorithms for learning diverse manipulation tasks.
 - Developed a data collection pipeline utilizing a haptic device.

PUBLICATIONS

- [C.1] Sunin Kim*, Jaewoon Kwon*, Taeyoon Lee*, Younghyo Park* and Julien Perez: **Safety-aware unsupervised skill discovery**. *International Conference on Robotics and Automation (ICRA), 2023* * Equal Contribution, listed in alphabetical order.
- [J.1] Sunin Kim, HyunJun Jo and Jae-Bok Song: **Object manipulation system based on image-based reinforcement learning**. *Intelligent Service Robotics (ISR), 2022*.
- [C.2] Sunin Kim, HyunJun Jo and Jae-Bok Song. **Generalized Object Manipulation Based on Keypoints Detection Network**. *The Korean Society of Mechanical Engineers(KSME) Conference, 2021*.
- [C.3] Sunin Kim and Jae-Bok Song. **Pose Estimation using RGB-based Deep Learning and Point Cloud-based correction**. *The Korean Society of Mechanical Engineers(KSME) Conference, 2020*.

SKILLS

- **Programming Languages:** Python, C++
- **Libraries/Frameworks:** PyTorch, NVIDIA Isaac Sim/Lab, MuJoCo, ROS2

AWARDS

- **Best Paper Award**

Jul. 2021

18th International Conference on Ubiquitous Robots

- [J.1] Object manipulation system based on image-based reinforcement learning